Figure 1: Plot of i ($\sqrt{-1}$)

0.01, and then let's plot the two curves in blue and red:

```
$ octave -qf
octave:1> th=0:0.01:pi;
octave:2> plot(th, exp(i*th), "b";exp, th,
cos(th)+i*sin(th), "r";cs;")
octave:3>
```

Figure 2 shows the two curves coinciding exactly with each other.

Equipped with i , let's play around with more puzzles

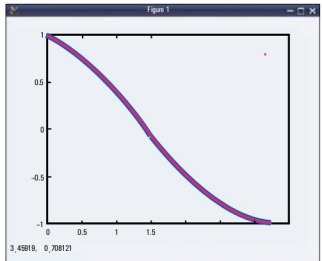


Figure 2: Plot of Euler's equality

in the following issues of OSFY; so get ready with the Octave controls. **END** 

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```
of file
cd $HADOOP_HOME
./bin/hadoop jar hadoop-examples-1.0.4.jar wordcount /user/
hduser/input /user/hduser/output
./bin/hadoop dfs -cat /user/hduser/output/*
./bin/stop-all.sh
```

The first line in the code starts the required services for Hadoop. The `jps` command lets you query all the Java Virtual Machines running on your system. You should see the following services running on your system.

1. NameNode
2. DataNode
3. Secondary NameNode
4. Job Tracker
5. Task Tracker
6. Jps

If any of the services listed above is not running, it means that your Hadoop could not start properly. In Line 3, create a local folder to be copied on to HDFS. In Line 4, make a text file and fill it with some random text. In Line 6, list the contents of the copied file. In Line 8, run the Hadoop word

count example that comes with the distribution. In Line 9, list the output of the generated result. Finally, in Line 10, stop all the Hadoop services.

This article has covered various aspects of big data, analytics and Hadoop. I have primarily focused on the Hadoop architecture and pointed out the loopholes of Hadoop in terms of fault tolerance and recovery. We have also specifically seen how NameNode and Job Tracker are bottlenecks in the system. They act as a single point of failure for the whole system. Many of the Hadoop distributions try to solve this problem of fault tolerance and recovery that is found in Hadoop Core. However, implementation of the fault tolerance and recovery algorithm creates performance issues. The thrust of the research in this field is to mitigate the performance and reliability issues to get the best of both worlds. **END** 

By: Saurabh Jha

The author is an open source enthusiast who loves to create awareness about free and open source software and technologies. His interest includes high performance computing in heterogeneous environments. He can be reached at his website : <http://saurabhjha.in>